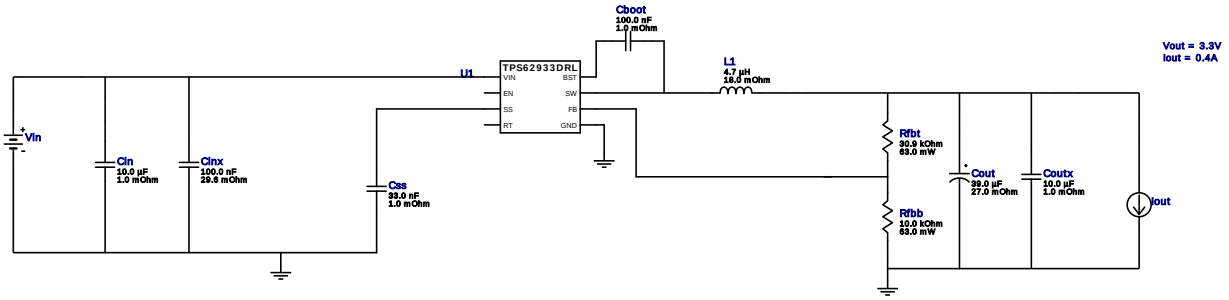
WEBENCH ® Design Report

VinMin = 8.0V
VinMax = 30.0V
Vout = 3.3V
Iout = 0.4A

Device = TPS62933DRLR
Topology = Buck
Created = 2025-03-14 23:37:00.799
BOM Cost = $1.34
BOM Count = 10
Total Pd = 0.22W

Design : 19 TPS62933DRLR
TPS62933DRLR 8V-30V to 3.30V @ 0.4A



Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cboot	MuRata	GRM155R71A104KA01D Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
Cin	TDK	C3216X5R1H106K160AB Series= X5R	Cap= 10.0 uF ESR= 1.0 mOhm VDC= 50.0 V IRMS= 4.9 A	1	$0.24	1206_180 11 mm²
Cinx	TDK	CGA3E2X7R1H104K080AA Series= X7R	Cap= 100.0 nF ESR= 29.6 mOhm VDC= 50.0 V IRMS= 971.99 mA	1	$0.01	0603 5 mm²
Cout	Panasonic	16SVPC39MV Series= SVPC	Cap= 39.0 uF ESR= 27.0 mOhm VDC= 16.0 V IRMS= 2.35 A	1	$0.44	SM_RADIAL_5MM 58 mm²
Coutx	MuRata	GRM155R60J106ME15D Series= X5R	Cap= 10.0 uF ESR= 1.0 mOhm VDC= 6.3 V IRMS= 3.52 A	1	$0.03	0402 3 mm²
Css	MuRata	GRM155R71A333KA01D Series= X7R	Cap= 33.0 nF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
L1	NIC Components	NPI31W4R7MTRF	L= 4.7 µH 18.0 mOhm	1	$0.23	IND_NPI31W 172 mm²
Rfbb	Vishay-Dale	CRCW040210K0FKED Series= CRCW..e3	Res= 10.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Rfbt	Vishay-Dale	CRCW040230K9FKED Series= CRCW..e3	Res= 30.9 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
U1	Texas Instruments	TPS62933DRLR	Switcher	1	$0.35	DRL0008A-MFG 9 mm²

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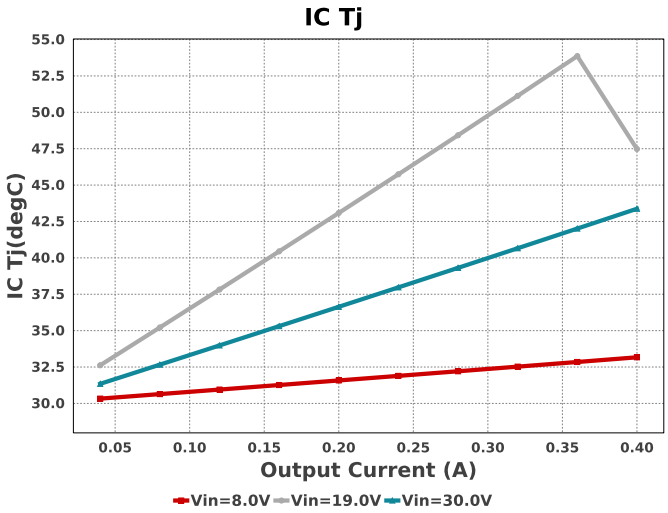

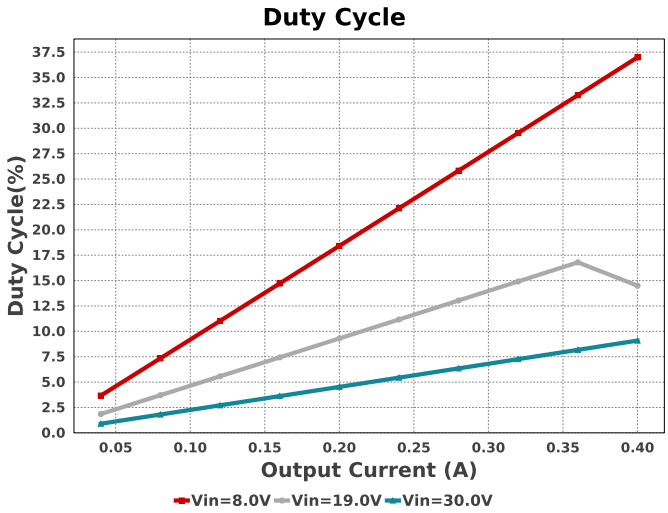

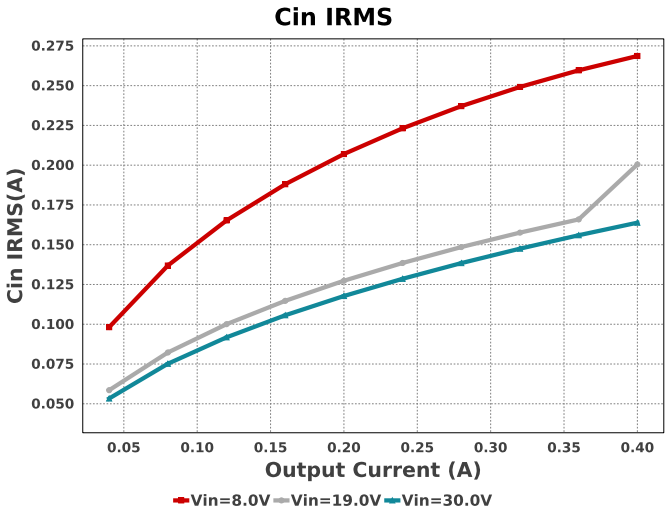

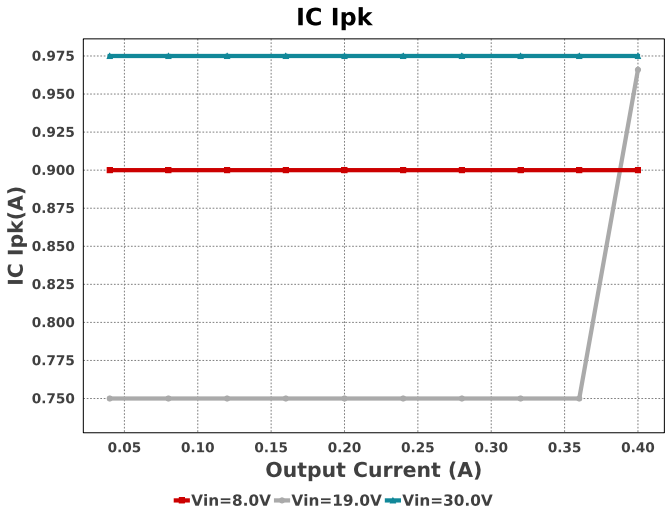

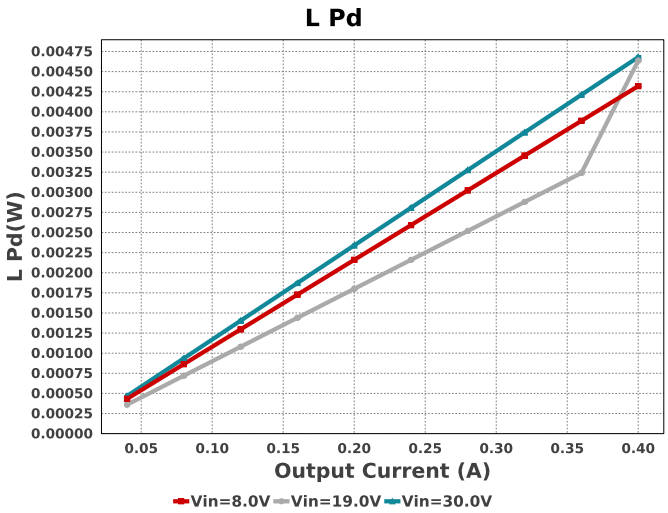

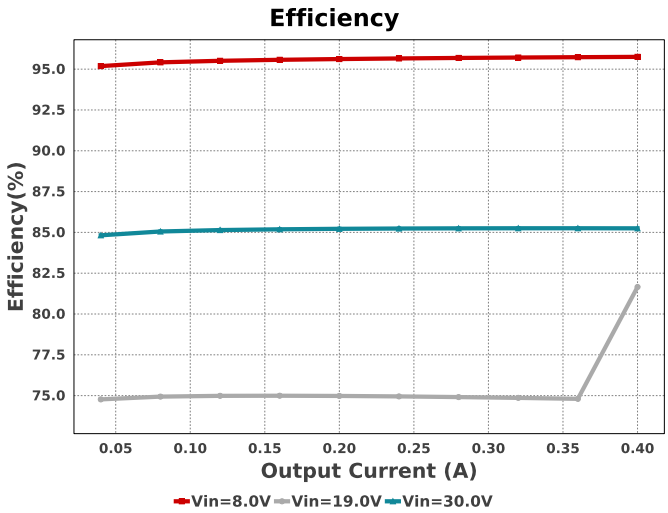

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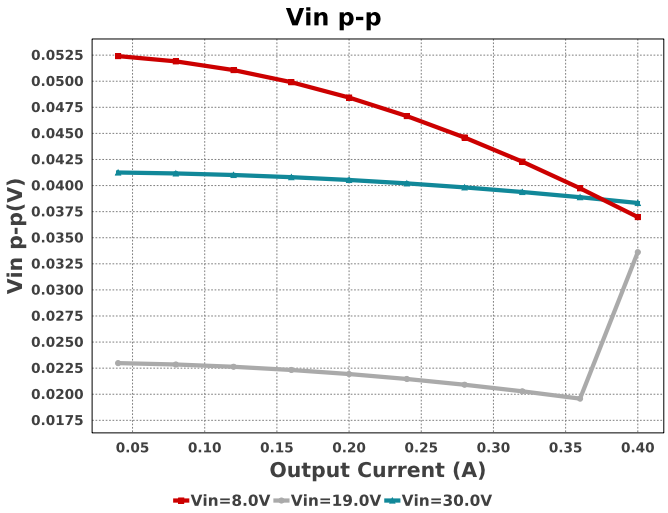

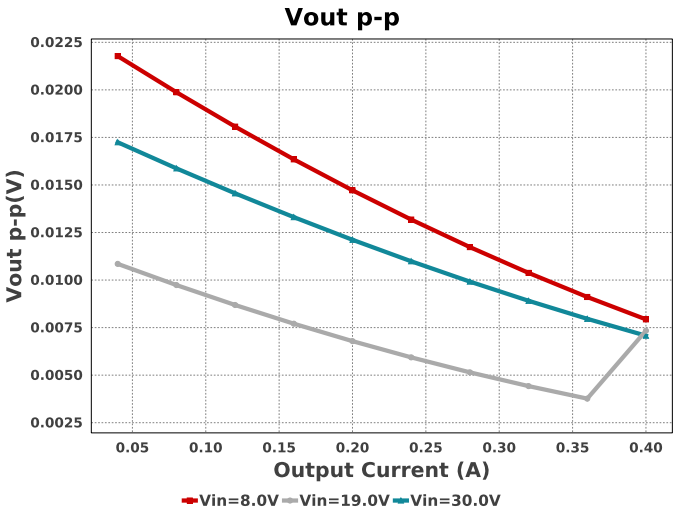

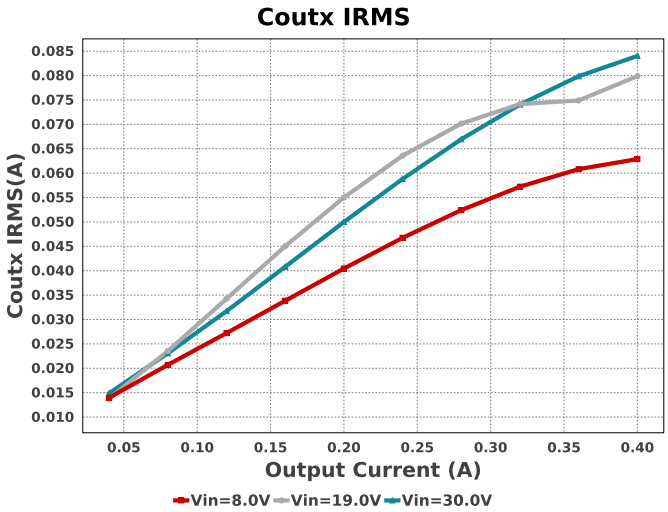

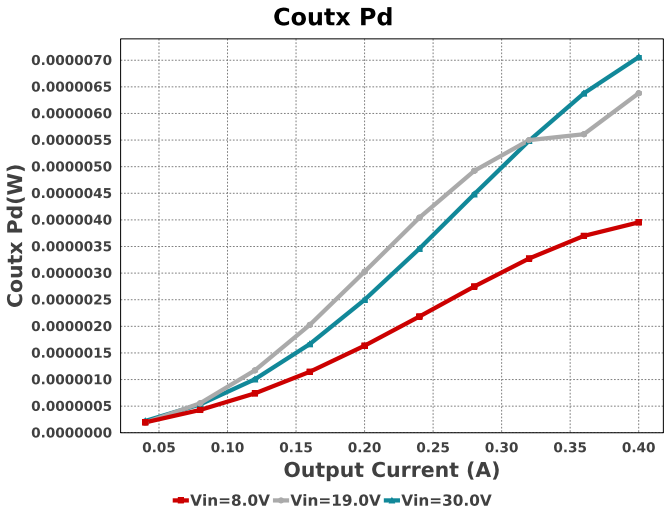

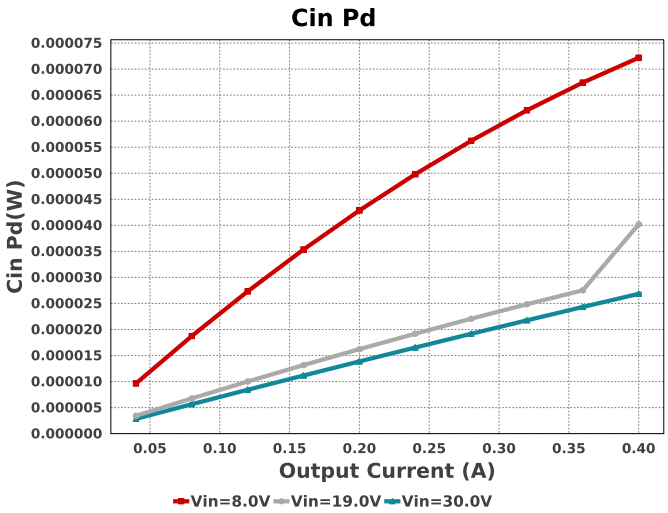

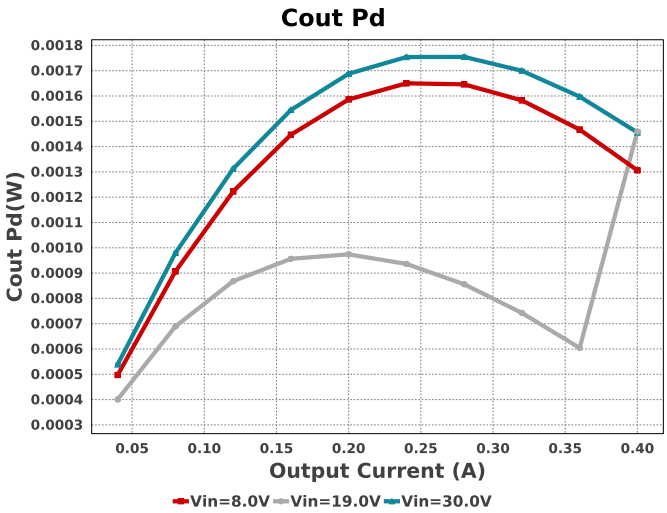

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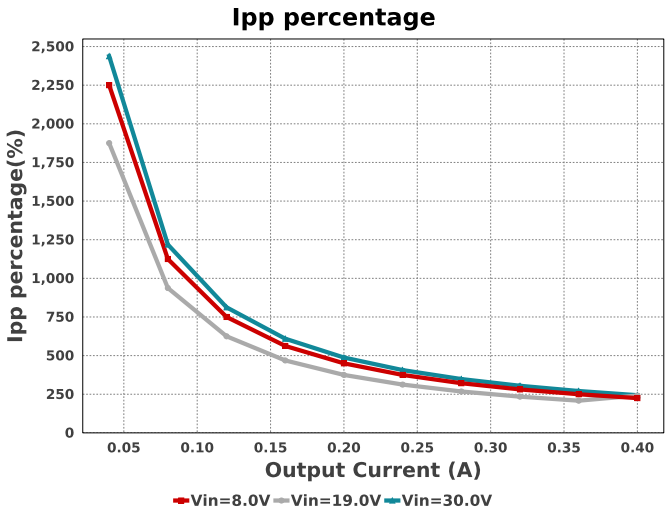

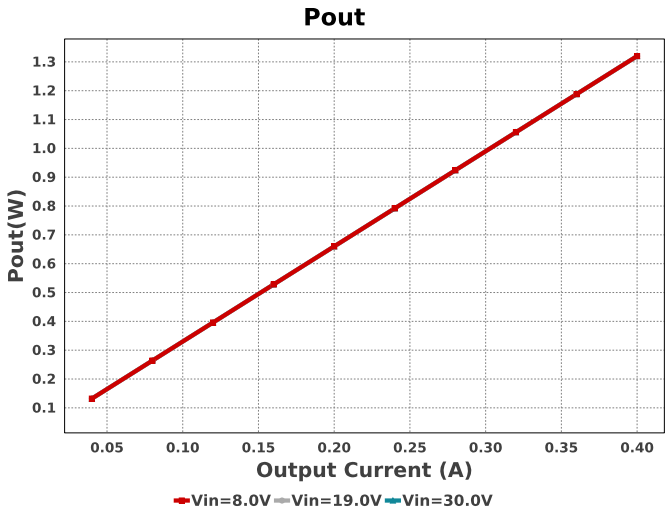

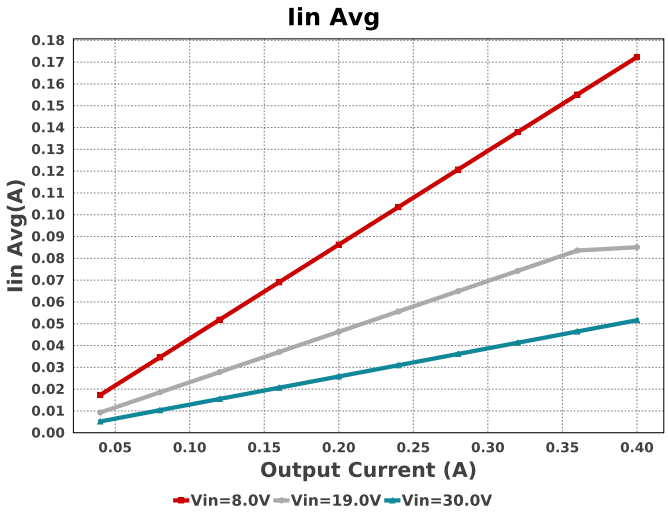

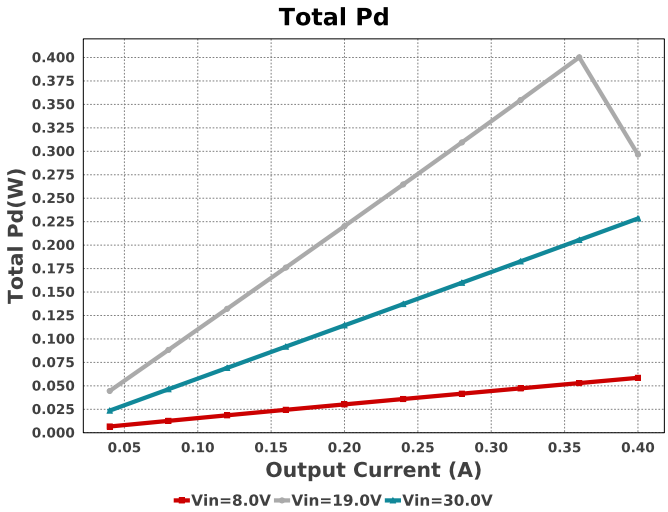

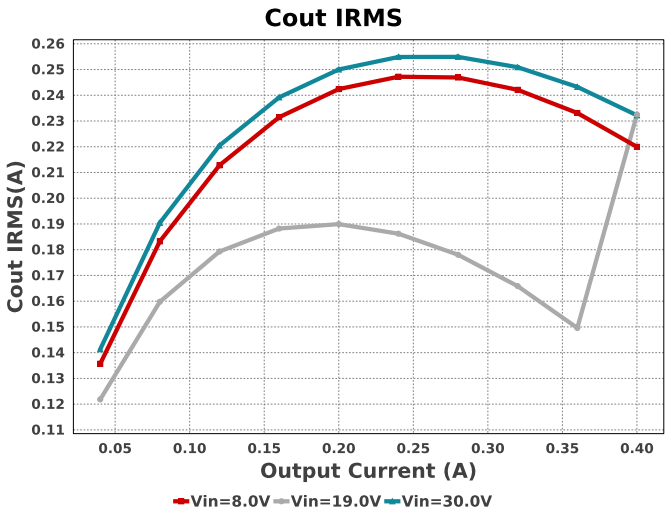

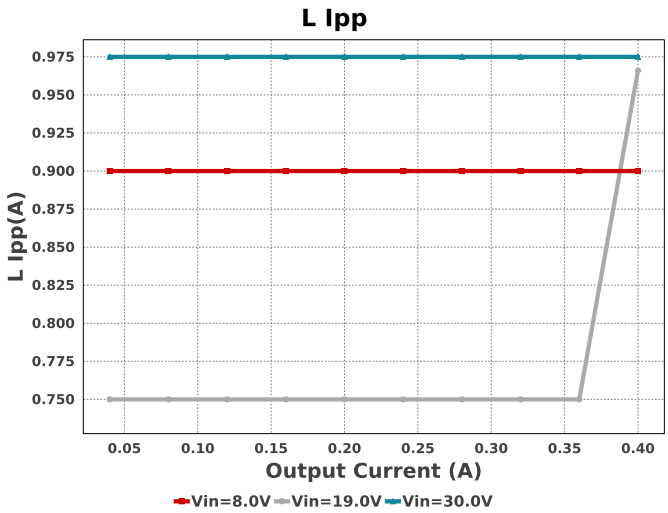

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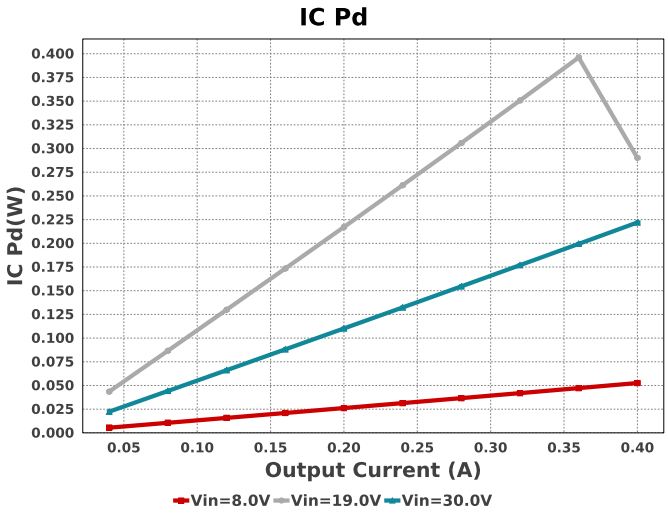

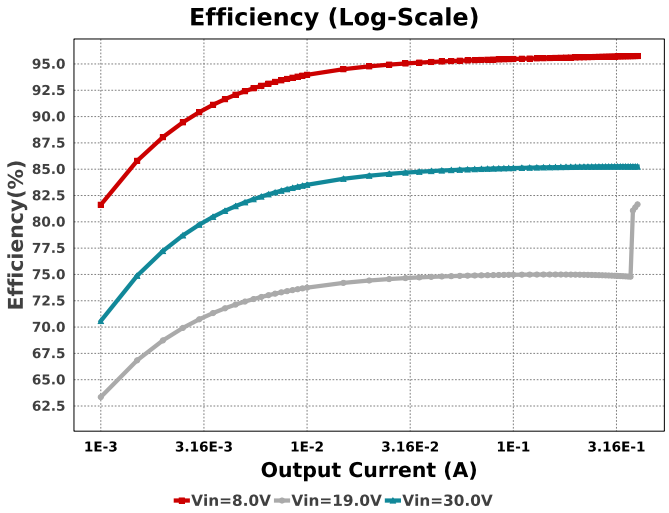

Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	166.335 mA	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	27.667 µW	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	243.933 mA	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	1.607 mW	Capacitor	Output capacitor power dissipation
5.	Coutx IRMS	83.804 mA	Capacitor	Output capacitor_x RMS ripple current
6.	Coutx Pd	7.023 µW	Capacitor	Output capacitor_x power loss
7.	IC Ipk	1.003 A	IC	Peak switch current in IC
8.	IC Pd	215.47 mW	IC	IC power dissipation
9.	IC Tj	42.982 degC	IC	IC junction temperature
10.	IC Tolerance	16.0 mV	IC	IC Feedback Tolerance
11.	ICThetaJA Effective	60.25 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
12.	Iin Avg	51.406 mA	IC	Average input current
13.	Ipp percentage	250.698 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
14.	L Ipp	1.003 A	Inductor	Peak-to-peak inductor ripple current
15.	L Pd	4.813 mW	Inductor	Inductor power dissipation
16.	Cin Pd	27.667 µW	Power	Input capacitor power dissipation
17.	Cout Pd	1.607 mW	Power	Output capacitor power dissipation
18.	Coutx Pd	7.023 µW	Power	Output capacitor_x power loss
19.	IC Pd	215.47 mW	Power	IC power dissipation
20.	L Pd	4.813 mW	Power	Inductor power dissipation
21.	Total Pd	222.197 mW	Power	Total Power Dissipation
22.	BOM Count	10	System Information	Total Design BOM count
23.	Cross Freq	28.735 kHz	System Information	Bode plot crossover frequency
24.	Duty Cycle	8.84 %	System Information	Duty cycle
25.	Efficiency	85.592 %	System Information	Steady state efficiency
26.	FootPrint	270.0 mm^2	System Information	Total Foot Print Area of BOM components
27.	Frequency	500.0 kHz	System Information	Switching frequency
28.	Gain Marg	-26.318 dB	System Information	Bode Plot Gain Margin
29.	Inductor ripple current requirement used for Inductor selection	40.0 %	System Information	Custom Inductor ripple current (% of average inductor current) requirement used for Inductor selection
30.	Iout	400.0 mA	System Information	Iout operating point
31.	Iout transient step used for Cout calculations	200.0 mA	System Information	Custom Transient current step requirement that was used for Cout selection (A).
32.	Low Freq Gain	106.364 dB	System Information	Gain at 1Hz
33.	Mode	DCM	System Information	Conduction Mode
34.	Overshoot Value	664.335 µV	System Information	Theoretical Vout Overshoot Value
35.	Phase Marg	59.371 deg	System Information	Bode Plot Phase Margin

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#	Name	Value	Category	Description
36.	Pout	1.32 W	System Information	Total output power
37.	Total BOM	$1.34	System Information	Total BOM Cost
38.	Undershoot Value	8.385 mV	System Information	Theoretical Vout Undershoot Value
39.	Vin	30.0 V	System Information	Vin operating point
40.	Vin p-p	40.685 mV	System Information	Peak-to-peak input voltage
41.	Vout	3.3 V	System Information	Operational Output Voltage
42.	Vout Actual	3.272 V	System Information	Vout Actual calculated based on selected voltage divider resistors
43.	Vout Ripple requirement used for Cout calculations	1.0 %	System Information	Custom maximum output ripple requirement that was used for Cout selection(% of Vout).
44.	Vout Tolerance	3.557 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
45.	Vout p-p	7.709 mV	System Information	Peak-to-peak output ripple voltage
46.	Vout transient requirement used for Cout calculations	3.0 %	System Information	Custom Transient voltage change requirement that was used for Cout selection (% of Vout).

Design Inputs

Name	Value	Description
Iout	400.0 m	Maximum Output Current
SoftStart	5.0 ms	Soft Start Time (ms)
VinMax	30.0	Maximum input voltage
VinMin	8.0	Minimum input voltage
Vout	3.3	Output Voltage
base_pn	TPS62933	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature
UserFsw	500.0 k	Customer Selected Frequency

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WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

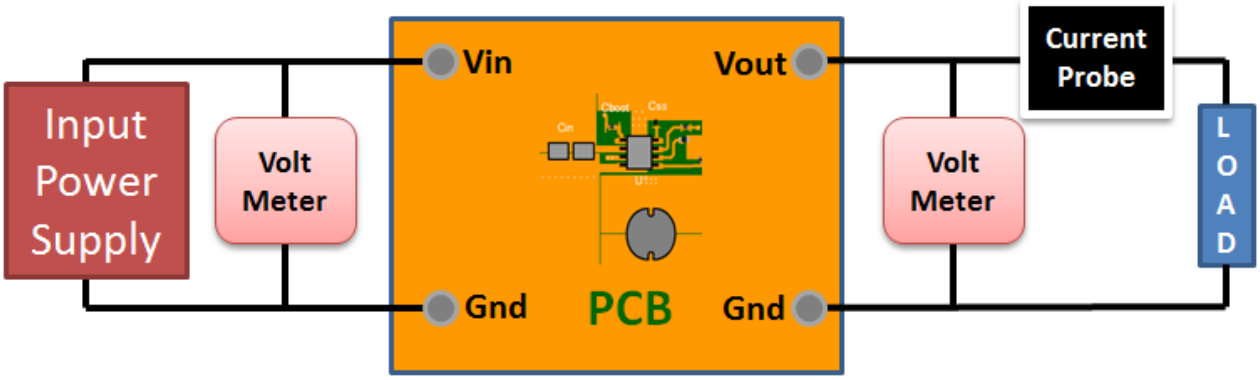
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 8.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.

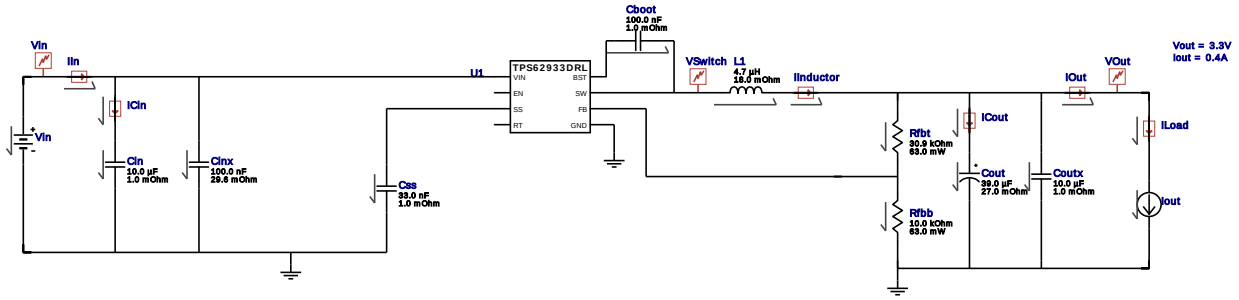


WEBENCH® Electrical Simulation Report

Design Id = 19

sim_id = 1

Simulation Type = Steady State



Simulation Parameters

#	Name	Parameter Name	Description	Values
1.	L1	IC	Initial Current	0.4 A
2.	Cout	IC	Vout	3.3
3.	Css	IC	no description	0.8 V
4.	Iout	I	Output Current	0.4 A

Design Assistance

1. Master key : D9474AC033CF2E09[v1]

2. **TPS62933** Product Folder : http://www.ti.com/product/TPS62933 : contains the data sheet and other resources.

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